

Why joint training may fail to acquire useful scales

Readout learning changes the residual that drives slope learning. The companion brief, *Gamma controls readout learning speed*, establishes why scale matters. Here we ask why joint training often fails to acquire it. Rate interventions support competition between readout and geometry; an early theorem captures signal depletion; the current approach asks which remaining errors can still drive slope growth. GD and Adam help separate gradient activity from sustained outward motion. A quantitative long-time prediction is still open.

Notation. Parameter-vector norms are Euclidean; residual norms use the stated data measure.

a_j, b_j, w_j, d	Hidden slope, hidden bias, readout weight, and output bias.
$\gamma_j = a_j , \bar{\gamma}$	Individual slope magnitude and its population mean.
$r = f - y, L = \ r\ _m^2/2$	Residual and half-MSE; $\ r\ _m^2 = m^{-1} \sum_i r(x_i)^2$.
$\kappa, t = \eta n$	Readout/geometry rate ratio and flow clock for GD update n .
$e_C, e_H, \mu_a^{\text{eff}}$	Mean/linear error, remaining error, and effective slope sensitivity (Section 3).

1. Empirical motivation: changing the race changes gamma

The jointly trained model is

$$f(x) = d + \sum_{j=1}^W w_j \tanh(a_j x + b_j). \quad (1)$$

Our first mechanistic evidence was that reparameterizing readout versus geometry changed gamma acquisition. A fixed scaling $\theta = Du$ makes a GD step in u induce $\Delta\theta = -\eta DD^T \nabla_{\theta} L$. For example, training signed slopes in grid units ha_j , with fixed reference spacing h , multiplies the physical slope step by h^{-2} . These changes alter the relative mobility of readout and geometry. The subsequent direct intervention isolates the rate ratio: keep slope and hidden-bias rate η fixed, and vary the readout and output-bias rate $\kappa\eta$, from exactly the same random initial parameters [1].

Why measure the mean and linear trend? At small initialization, $\tanh(a_j x + b_j) \approx a_j x + b_j$, so the network mainly represents a constant plus a linear trend. We call the projection of the residual onto these two functions the *coarse error*; the remaining non-affine error is the *fine error*. This is a definition of what we measure, not an assumption that training has already fitted it. Figure 1 tests whether faster readout fitting leaves less slope growth. Each horizontal position is a different rate ratio, not a training time.

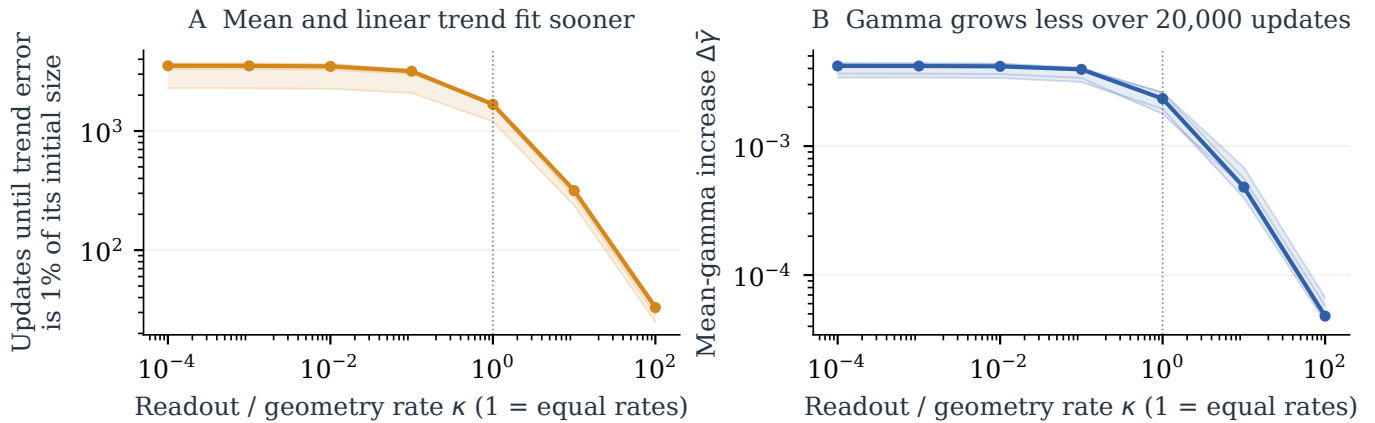


Figure 1: Faster readout fits the mean and linear trend sooner, while gamma grows less. A: updates until the coarse-error norm reaches 1% of its initial value. B: net change in mean γ after the same 20,000 updates. The illustrative target combines a constant, a linear trend, and a degree-nine polynomial component. All parameters train simultaneously: $W = 177, 2,048$ training points, geometry rate 0.002, five paired seeds. Bold lines are medians; faint lines and shading show the seeds.

The trend is consistent with a race: readout adaptation can reduce the error that would otherwise move the slopes, before much gamma growth accumulates. Fine error remains large, and the growth contrast is also present when runs are compared at the same coarse-error threshold. This is mechanistic evidence for competition. Its direction varies with the target, so it does not yet give a universal rule for choosing the relative rates.

2. First proof idea: quantify early signal depletion

The first proof asks whether readout fitting removes an initially available slope signal while appreciable error remains. A small gradient alone does not answer this: it might simply reflect successful fitting. We therefore study

$$R(t) = \|r(t)\|, \quad g_a = \nabla_a L, \quad \Xi(t) = \frac{\|g_a(t)\|}{R(t)}. \quad (2)$$

This is the current slope gradient per unit residual, including any signal regenerated during training [2].

The theorem’s setting. Use equal-rate gradient flow, continuous $L^2(dx/2)$ loss on $[-1, 1]$, no output bias, and *zero initial readout*. Hidden slopes and biases are independent uniforms on $[-\sqrt{6/(W+1)}, \sqrt{6/(W+1)}]$. The fixed target y is centered, normalized, and entire, with $\beta = 3\langle y, x \rangle \neq 0$ and $\|y - \beta x\| > 0$. For each fixed $H_0 > 3$, at sufficiently large width and with arbitrarily high fixed probability, signal first forms at a width-independent time t_f , then obeys

$$\Xi(t_f) \geq s_y > 0, \quad \Xi(t) \leq C_{y, H_0}(e^{-t/3} + W^{-1}), \quad 0 \leq t \leq H_0 \log W. \quad (3)$$

In particular, $\Xi = O(W^{-1})$ throughout $[3 \log W, H_0 \log W]$, even though the target has a non-affine component.

Why the proof works. At small preactivations, $\tanh$ is approximately affine. Its exact slope gradient can be written

$$(g_a)_j = w_j \langle r, x \rangle - w_j \langle r, x \tanh^2(a_j x + b_j) \rangle. \quad (4)$$

The coupled affine reference describes fitting the constant and linear moments. A parameter comparison and nonlinear remainder estimates bound both their regeneration and the second term in (4). Thus the theorem bounds the total surviving signal, rather than merely showing that the initial coarse residual disappeared.

Where the readout enters. Let $q = (a, b)$ and let H_{aI} be the corresponding loss-Hessian block. Where $\|g_a\|, R > 0$, equal-rate flow gives the exact identity

$$-\frac{d}{dt} \log \Xi = D_w + D_q, \quad D_I = \frac{g_a^T H_{aI} g_I}{\|g_a\|^2} - \frac{\|g_I\|^2}{R^2}. \quad (5)$$

On the shorter attribution window from t_f to $t_f + \rho \log W / \nu_y$, with $0 < \rho < 1/2$ and $\nu_y = 2\sqrt{1 + \beta^2}/3$, the readout’s share of the leading logarithmic decline tends in probability to

$$\pi_y = \frac{1}{2} + \frac{1}{2\sqrt{1 + \beta^2}} \in (3/4, 1). \quad (6)$$

This is a signed attribution along the joint trajectory. It says more than three quarters of that leading decline is attributable to readout adaptation; it is not three quarters of all possible gamma growth, or a frozen-readout counterfactual. Individual contributions can later become negative.

Why this was not yet quantitatively predictive. The proof supplies width and time scaling, but its conservative constants do not give tight bounds at our experimental widths. Uniform remainder estimates can be much larger than the surviving signal they bound. The asymptotic readout share quantifies a mechanism, but does not predict how far gamma moves, when useful scales are reached, or whether nonlinear forcing later recovers. The guarantee also covers an early logarithmic-time window and a zero-readout law, whereas the long D34 trajectories start with random nonzero readouts. Accurate cubic or quintic reference forecasts help, but measured agreement does not certify a uniformly small error envelope.

The proof establishes an early mechanism. The next task is to explain the force that remains, including its subsequent recovery, on the actual training trajectory. That motivates resolving the gradient-flow equations more directly.

3. Current approach: identify the force that still moves slopes

Return to the empirical model (1), with all four blocks trained, random nonzero readouts, and equal physical rates. Its slope equation is

$$\dot{a}_j = -(g_a)_j = -w_j \langle r(x) x \operatorname{sech}^2(a_j x + b_j) \rangle_m. \quad (7)$$

Residual size alone cannot determine this velocity: the residual must align with each current slope tangent, and the readout weight multiplies that tangent. We now separate the contribution associated with the mean and linear trend from the force supplied by the remaining error [3].

Resolve the residual into coarse and fine components. Choose a fixed empirical orthonormal basis $q_k(x)$ on the training points, with $q_0 = 1$, $q_1 = x/\sqrt{\langle x^2 \rangle_m}$ on the symmetric grid. Set $e_k = \langle q_k r \rangle_m$, $e_C = (e_0, e_1)^T$, and let e_H contain the *complete* remaining complement. Define the modal Jacobian block $(J_I)_{kj} = \langle q_k \partial f / \partial \theta_{I,j} \rangle_m$ for $I \in \{a, b, w, d\}$. Then $g_I = J_I^T e$, and equal-rate flow gives

$$\dot{e} = -K e, \quad K = \sum_I J_I J_I^T, \quad \begin{aligned} \dot{e}_C &= -K_{CC} e_C - K_{CH} e_H, \\ \dot{e}_H &= -K_{HC} e_C - K_{HH} e_H. \end{aligned} \quad (8)$$

All parameter blocks contribute to K . The coarse/fine split concerns output error, rather than a split between readout and geometry parameters.

Coarse equilibrium is generally nonzero. If K_{CC} is invertible, hold its current blocks and e_H fixed to find the instantaneous coarse balance:

$$B = K_{CC}^{-1} K_{CH}, \quad e_C^{\text{bal}} = -B e_H, \quad z_C = e_C + B e_H. \quad (9)$$

Here z_C measures disagreement with that moving balance. Substituting $e_C = -B e_H + z_C$ into the slope gradient and fine-residual equation gives

$$\boxed{g_a = J_{a,C}^T z_C + T_a e_H, \quad \dot{e}_H = -S e_H - K_{HC} z_C}, \quad (10)$$

$$T_a = J_{a,H}^T - J_{a,C}^T B, \quad S = K_{HH} - K_{HC} B. \quad (11)$$

The first force in (10) comes from imperfect tracking; the second survives at coarse balance. Crucially, $T_a e_H$ includes the balanced coarse response $-J_{a,C}^T B e_H$. Setting $e_C = 0$ would omit it. “Fine” also includes generated low-degree nonlinear errors, not only the hardest target mode. These identities are exact for the full complement; a truncated modal calculation must retain its omitted-residual corrections.

Why focus on the effective fine force? Across the post-transient GD panel (13 targets, five seeds), $J_{a,C}^T z_C$ is small and $T_a e_H$ accounts for nearly the whole slope gradient [4]. Adam gives a useful contrast: coarse disagreement can be large without supplying positive net gamma growth (Figure 2). Thus the quantity to understand is the surviving force and its sustained outward direction, not residual size or instantaneous gradient magnitude alone.

Arriving at the effective sensitivity. For $R_H = \|e_H\| > 0$, let $u_H = e_H / R_H$ and define

$$G_a = T_a^T T_a, \quad \boxed{\mu_a^{\text{eff}} = u_H^T G_a u_H = \frac{\|T_a e_H\|^2}{\|e_H\|^2}, \quad \|T_a e_H\| = R_H \sqrt{\mu_a^{\text{eff}}}}. \quad (12)$$

This is a squared, direction-dependent sensitivity of slopes to the current fine residual. It separates *how much error remains* from *how strongly that error can move the slopes*. It also incorporates interference between direct fine forcing and the induced coarse response.

Large gradient activity need not acquire scale

In the mixed-sine example, coarse disagreement is small in GD but comparable to the full slope gradient in Adam. Yet it supplies no positive signed mean-gamma growth in any of the five Adam seeds shown.

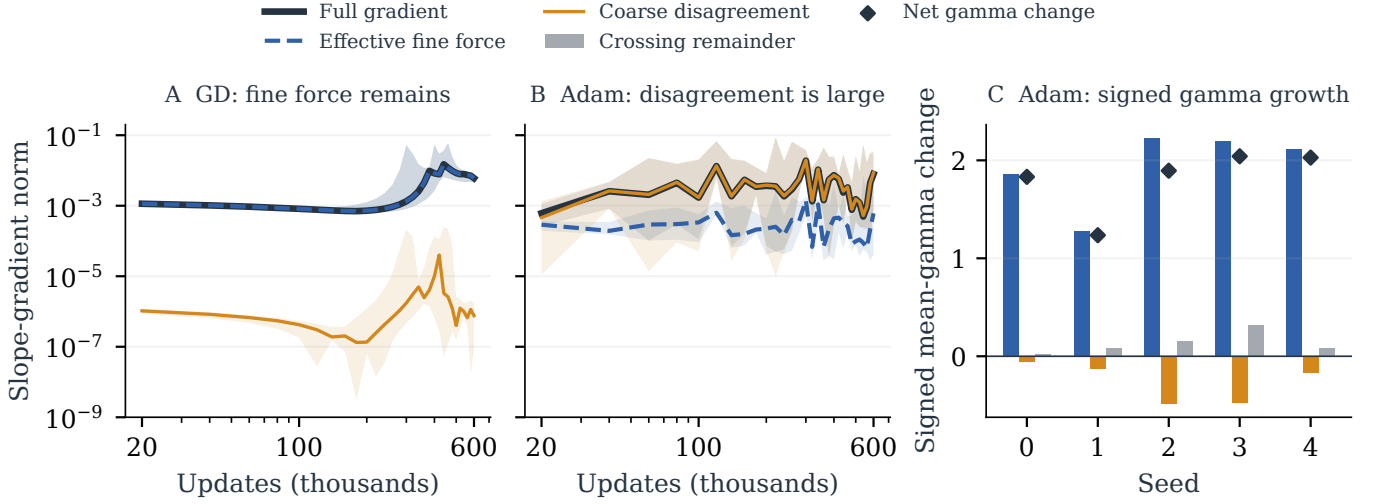


Figure 2: Adam’s large coarse-disagreement force does not imply outward growth. A–B: gradient norms on each optimizer’s trajectory, with a shared vertical scale; lines are five-seed medians, shading spans seeds. C: signed Adam contributions accumulated at every update; diamonds show net growth and gray accounts for zero crossings. Orange is negative despite its large norm in B. A normalized sum of three sine waves; $W = 177$, 2,048 points, rate 0.002, updates 20,000–600,000 [4].

For Adam, the test must use the actual updates. Split the first-moment history by channel and apply the *same actual second-moment denominator* to both. Then $\Delta a_n = \Delta a_n^{\text{eff}} + \Delta a_n^{\text{track}}$, with exact accounting

$$D_k = \sum_n \frac{\text{sign}(a_n)^T \Delta a_n^k}{W}, \quad \Delta \bar{\gamma} = D_{\text{eff}} + D_{\text{track}} + X. \quad (13)$$

Here $X \geq 0$ is the absolute-value remainder at zero crossings. Coarse activity mostly cancels or points inward, while the fine channel supplies positive drift. Tracking can contribute positively on other targets; this accounting does not establish what removing it would do. Large disagreement also remains when the instantaneous balance uses Adam’s metric [4].

Back in gradient flow, the race has an explicit dynamical form. Write $F_a = T_a e_H$. Differentiating the force gives

$$\dot{F}_a = \underbrace{\dot{T}_a e_H}_{\text{changing sensitivity map}} - \underbrace{T_a \dot{S} e_H}_{\text{evolving fine residual}} - \underbrace{T_a K_{HC} z_C}_{\text{tracking correction}}. \quad (14)$$

Readout evolution changes both T_a and S : coupling can improve while the driving residual depletes or rotates. Both weakening and renewed forcing remain possible.

To connect sensitivity to acquisition, define the signed outward alignment $\chi_a = -\text{sign}(a)^T F_a / (\sqrt{W} \|F_a\|)$ where $F_a \neq 0$. Away from slope zero crossings, the exact flow identity is

$$\dot{\gamma} = \frac{R_H \sqrt{\mu_a^{\text{eff}}} \chi_a}{\sqrt{W}} - \frac{\text{sign}(a)^T J_{a,C}^T z_C}{W}, \quad |\chi_a| \leq 1. \quad (15)$$

Sensitivity measures available force; alignment determines outward growth. A population claim also needs its distribution across neurons. Adam requires the filtered, preconditioned direction in (13).

Working hypothesis. Readout–geometry feedback may weaken sensitivity or deplete useful residual directions before enough outward motion accumulates. Controlling $R_H \sqrt{\mu_a^{\text{eff}}}$ and its alignment could predict acquisition time or bound attainable motion. This connects the ODEs directly to gamma growth; the missing proof is control of that feedback over time.

Sources and reproduction. [1] D34 rate intervention; [2] Early-depletion theory, 20 September 2026; [3] Coupled-force theory; [4] Expanded GD/Adam force study. The companion motivation is [Gamma controls readout learning speed](#). Figures replot existing fp64 runs. Inputs, values, checks, and hashes are in the [evidence record](#). Figure and PDF rebuild commands are in the LaTeX source. Links resolve locally.